

On the Role of MRI Sequences in Cross-Dataset Generalization for Brain Tumor Segmentation

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Abstract—Brain tumor segmentation in magnetic resonance imaging (MRI) is a critical task for diagnosis and treatment planning. Despite the success of deep learning architectures such as U-Net and its variants, performance degradation across datasets remains a major challenge, particularly under domain shift and limited annotated data. To address this issue, this study systematically evaluates how individual MRI sequences influence model robustness across two well-known datasets. A ResUNet-based framework is employed, where each modality is trained independently to isolate its effect under a controlled cross-dataset evaluation protocol with tumor size stratification, without target-domain training or with limited domain adaptation. Results show that the T2f/FLAIR sequence achieves the best cross-dataset performance, with Dice scores exceeding 75%. It consistently outperforms other modalities across most tumor size ranges, while multi-sequence training further improves performance. Additionally, even limited target-domain adaptation yields rapid initial gains, reducing the need for extensive annotations and costly retraining.

I. INTRODUCTION

Automatic brain tumor segmentation in magnetic resonance imaging (MRI) is a prominent application of deep learning in medical computer vision [1], [2], [3]. This task plays a critical role in diagnosis, surgical planning, and clinical follow-up, supporting clinicians in the analysis of imaging data. Advances in convolutional neural network architectures, especially U-Net and its variants, have achieved strong performance on benchmark datasets and competitions such as BraTS[4], [5], establishing them as a standard approach for brain lesion segmentation.

However, despite these advances, model performance often fails to generalize across datasets. Variations in acquisition protocols, scanners, patient populations, and image characteristics introduce domain shift, often leading to significant performance degradation on unseen data. This challenge, commonly referred to as cross-dataset generalization [6], is further exacerbated by the limited availability of annotated medical images, which constrains model robustness and real-world applicability. One of the key factors influencing cross-dataset generalization is the choice of MRI sequences used as model input, as each modality (T1n, T1c, T2w, and T2f/FLAIR) captures complementary tissue characteristics, differing in contrast, sensitivity to edema, and delineation of tumor boundaries [7], [8].

Unlike prior works that focus on maximizing performance through single-modality or multimodal fusion strategies, there is a lack of a systematic, analysis-driven perspective that isolates modality-specific effects under domain shift. To address this gap, this study systematically evaluates each modality independently, enabling a direct assessment of robustness and providing practical insights for scenarios with limited computational resources or incomplete multimodal data. Furthermore, the analysis of limited target-domain adaptation offers guidance on reducing annotation and retraining costs.

Based on the above discussion, some research questions arise:

- **(RQ1)** How does segmentation performance degrade across domains as a function of MRI modality?
- **(RQ2)** Which MRI sequences exhibit greater robustness in cross-dataset binary brain tumor segmentation?
- **(RQ3)** What is the relationship between target-domain annotation effort and performance improvement for different MRI modalities?

To address these questions, we conduct a systematic comparative analysis of cross-dataset generalization across MRI sequences for binary brain tumor segmentation (tumor vs. non-tumor). A standardized pipeline based on the widely used 2D ResUNet is employed, where each modality is trained and evaluated independently under a controlled protocol [6] to isolate its contribution to robustness. Additionally, we apply fine-tuning to the best-performing modality to assess the impact of limited target-domain data on generalization, with emphasis on a controlled and analysis-driven evaluation setting.

The contributions of this work are fourfold: (i) a controlled evaluation of modality-specific robustness under domain shift; (ii) quantification of the generalization gap across modalities; (iii) identification of the most robust MRI sequence for cross-dataset binary segmentation; and (iv) analysis of data-efficient domain adaptation, relating annotation budget to performance gains.

The remainder of this paper is organized as follows. Section II reviews related work and highlights the research gap. Section III describes the datasets, materials, and methodology. Section IV presents the experimental setup and evaluation metrics. Section V discusses the results. Finally, Section VI concludes the paper and outlines directions for future work.

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II. LITERATURE REVIEW

The literature on brain tumor segmentation in MRI is largely dominated by convolutional neural network-based approaches, particularly U-Net and its variants [1], [2]. Prior work has focused on architectural improvements and multimodal fusion strategies to enhance segmentation accuracy. Despite these advances, challenges related to domain shift and cross-dataset generalization remain insufficiently addressed.

Over the past decade, U-Net has been established as the baseline architecture for brain tumor segmentation due to its encoder-decoder design with skip connections [9]. However, its performance degrades in heterogeneous imaging scenarios, where variability in tumor morphology and intensity limits generalization [10].

To mitigate these limitations, recent variants such as ResUNet, ResUNet+, and nnU-Net have emerged, incorporating residual blocks, attention mechanisms, and automatic parameter adjustment [11], [12], [13]. The ResUNet model, in particular, combines the structural advantages of U-Net with the depth of residual networks, enabling more efficient training and the preservation of details in complex tumor regions [11]. Recent results indicate substantial gains: Mangayarkarasi et al. [12] reported an accuracy of 93.75% with ResUNet for brain tumor segmentation in LGG MRI dataset, while Metlek and Çetiner [13] showed that ResUNet+, with spatial mechanism, increased the mean DICE by up to 5.58% in whole tumor segmentation when compared with other SOTA methods. In parallel, nnU-Net has shown consistent performance by automating architectural and preprocessing decisions, reducing manual tuning and improving reproducibility [11].

In addition to architectural improvements, the literature has extensively explored multimodal fusion of MRI sequences (T1, T1c, T2, and T2f/FLAIR), aiming to integrate complementary contrast and structural information. Fusion strategies have been widely explored to balance computational cost while achieving increasingly better results; works such as those by Yin et al. [7] and Xing et al. [8] demonstrate that the combination of modalities increases segmentation accuracy and robustness, reaching Dice coefficients above 90% when compared with the isolated use of a single sequence. Three-dimensional models (e.g., ResUNet variations with dual or mirrored encoders) tend to better capture volumetric continuity and inter-slice consistency, although at the cost of greater computational demand and increased sensitivity to class imbalance [8].

Although existing approaches achieve near-human performance in the source domain, generalization remains a critical challenge. Variations in acquisition protocols, scanners, intensity distributions, and class imbalance degrade performance on unseen domains. Zhang et al. [6] demonstrate that even minor domain shifts can significantly impact accuracy, motivating more robust learning strategies. In addition, preprocessing choices and loss functions further influence model behavior, particularly in challenging cases such as

small lesions [3], [2].

Although existing approaches achieve near-human performance in the source domain, their robustness across heterogeneous datasets remains limited. Variations in acquisition protocols, scanners, and intensity distributions introduce significant domain shifts, often leading to performance degradation on unseen data [6]. While multimodal MRI inputs are commonly used to mitigate these effects, the individual contribution of each modality to cross-domain generalization has not been systematically quantified, particularly under constrained data or annotation settings.

III. PROBLEM STATEMENT

Despite the strong performance of deep learning-based methods in controlled settings, their generalization across datasets remains limited due to domain shift induced by variations in acquisition protocols and data characteristics. A critical question is how different MRI modalities contribute to robustness under such conditions, particularly when annotation resources are limited.

To address this gap, this work conducts a controlled comparative analysis of cross-dataset robustness across MRI sequences using a standardized 2D ResUNet pipeline. By isolating each modality and evaluating performance under varying annotation regimes, we quantify their impact on generalization and data efficiency, providing practical guidance for modality selection in real-world and resource-constrained clinical scenarios.

The following section details the experimental design and methodological choices adopted to ensure a controlled and reproducible evaluation. We first describe the datasets and their characteristics (Section III-A), followed by the partitioning and preprocessing pipeline applied uniformly across all experiments. Next, the 2D ResUNet architecture and training configuration are presented, including loss functions and optimization strategy. Finally, we outline the evaluation

A. Datasets

The Brain Tumor Segmentation Challenge (BraTS) datasets from 2020 and 2024 [4], [5] were selected due to their widespread adoption as standardized benchmarks and their complementary characteristics for evaluating model robustness. BraTS 2020 comprises 369 multimodal MRI studies and serves as a well-established, controlled benchmark for method comparison. In contrast, BraTS 2024 extends this scale to a substantially larger and more heterogeneous cohort, totaling approximately 1350 cases, reflecting recent advances in data collection and acquisition diversity. Both datasets provide four MRI sequences (T1n, T1c, T2w, and T2f/FLAIR) along with expert-annotated segmentation masks for tumor subregions, as illustrated in Figures 1 and 2. While annotation protocols remain consistent across datasets, variations in tumor morphology, inter-patient characteristics, and acquisition settings (e.g., scanners and imaging protocols) introduce realistic domain shifts. This combination enables a controlled experimental setup in which labeling consistency is preserved while data heterogeneity

is increased, making these datasets particularly suitable for evaluating cross-dataset generalization and modality-specific robustness. Furthermore, their standardized preprocessing pipelines support fair comparative analysis while maintaining clinically relevant complexity. To ensure reproducibility, the original patient-wise organization and naming conventions were preserved whenever available.

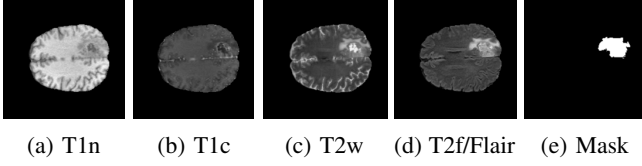


Fig. 1: Examples of BRATS (2020) images.

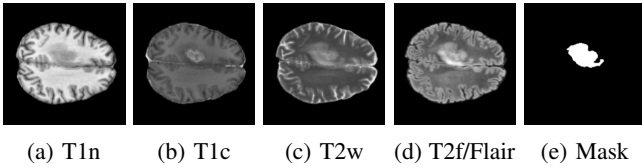


Fig. 2: Examples of BRATS (2024) images.

Data preparation followed a patient-level holdout strategy (70% training, 15% validation, 15% testing) to prevent data leakage and ensure mutually exclusive splits, avoiding that slices from the same patient appear across different subsets. All images were resized to 240×240 and normalized to the range $[0, 1]$, applying an identical preprocessing pipeline across datasets. For each MRI modality, 2D slice-based subsets containing only tumor-present slices were extracted to focus learning on informative regions and reduce computational cost. Multiclass annotations were converted into binary masks (tumor vs. non-tumor) to align with the task formulation. Data augmentation, including 90° rotations and horizontal/vertical flips, was applied exclusively to the training set. More aggressive transformations (e.g., intensity scaling, brightness or contrast jitter, and geometric distortions) were deliberately avoided, as they may alter anatomical structures or distort tumor appearance, potentially introducing unrealistic patterns and compromising clinical validity in medical imaging scenarios.

B. Architecture

The adopted architecture is a 2D ResUNet, which extends the conventional U-Net by integrating residual learning to improve feature representation and training stability [11], [12], [13], as illustrated in Figure 3. The model follows an encoder-decoder paradigm, where the encoder is based on a pretrained ResNet-34 backbone with channel progression (64, 128, 256, 512), enabling deep hierarchical feature extraction through identity mappings that facilitate gradient propagation. Multi-scale contextual features are progressively captured in the encoder, while skip connections preserve fine-grained spatial information by linking corresponding encoder and decoder stages. The decoder consists

of four upsampling stages ($512 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 64$), each combining transposed convolutions, feature concatenation, and double convolutional blocks (Conv-BatchNorm-ReLU with 3×3 kernels) for effective feature fusion and spatial refinement. Dropout regularization ($p = 0.2$) is incorporated within convolutional blocks to mitigate overfitting, particularly under limited-data and cross-dataset scenarios. The network processes single-channel inputs and produces a binary segmentation map through a final 1×1 convolution. This hybrid design combines the representational power of residual networks with the localization capability of U-Net, enabling accurate modeling of complex tumor morphology in MRI data.

Training Protocol and Hyperparameters. Models were trained in a supervised manner using a composite loss combining Dice Loss and Binary Cross-Entropy (BCE) [6], [8]. Experiments were conducted independently for each MRI modality. Training was performed for up to 80 epochs with a batch size of 32 and an initial learning rate of 1×10^{-3} , reduced adaptively based on validation performance. Early stopping (patience = 8) was applied using the mean Dice coefficient to prevent overfitting. The final model corresponds to the best validation Dice score.

IV. EXPERIMENTS AND DISCUSSION

To fairly assess the contribution of individual MRI modalities to cross-dataset generalization between BraTS 2020 [4] and BraTS 2024 [5], and to address the research questions defined in Section I, we organize the experimental analysis into two complementary perspectives. First, Section IV-A evaluates cross-domain generalization through baseline, cross-dataset, and multi-domain training protocols, quantifying performance under increasing levels of domain shift (RQ1/RQ2). Second, a domain adaptation strategy (Section IV-B) incorporates varying proportions of annotated target-domain data to assess the impact of annotation availability on performance and adaptation cost (RQ3). Additionally, performance is qualitatively analyzed across tumor sizes (Section IV-C), as overlap-based metrics (e.g., Dice) are more sensitive to small absolute errors in small lesions, potentially biasing comparisons and masking modality-specific performance trends.

The training, validation, and test samples were defined using a patient-level holdout strategy, as described in Section III-A. The baseline models¹ and training hyperparameters are detailed in Section III-B.

Performance was evaluated using overlap-based metrics widely adopted in medical image segmentation: the Dice coefficient and Intersection over Union (IoU) [14], [13]. Both quantify the agreement between predicted and reference masks. The Dice coefficient is defined as:

$$Dice = \frac{2TP}{2TP + FP + FN} \quad (1)$$

¹To ensure full reproducibility, all models, trained weights, source code, and experimental artifacts will be publicly released in a repository (link to be provided in the camera-ready version).

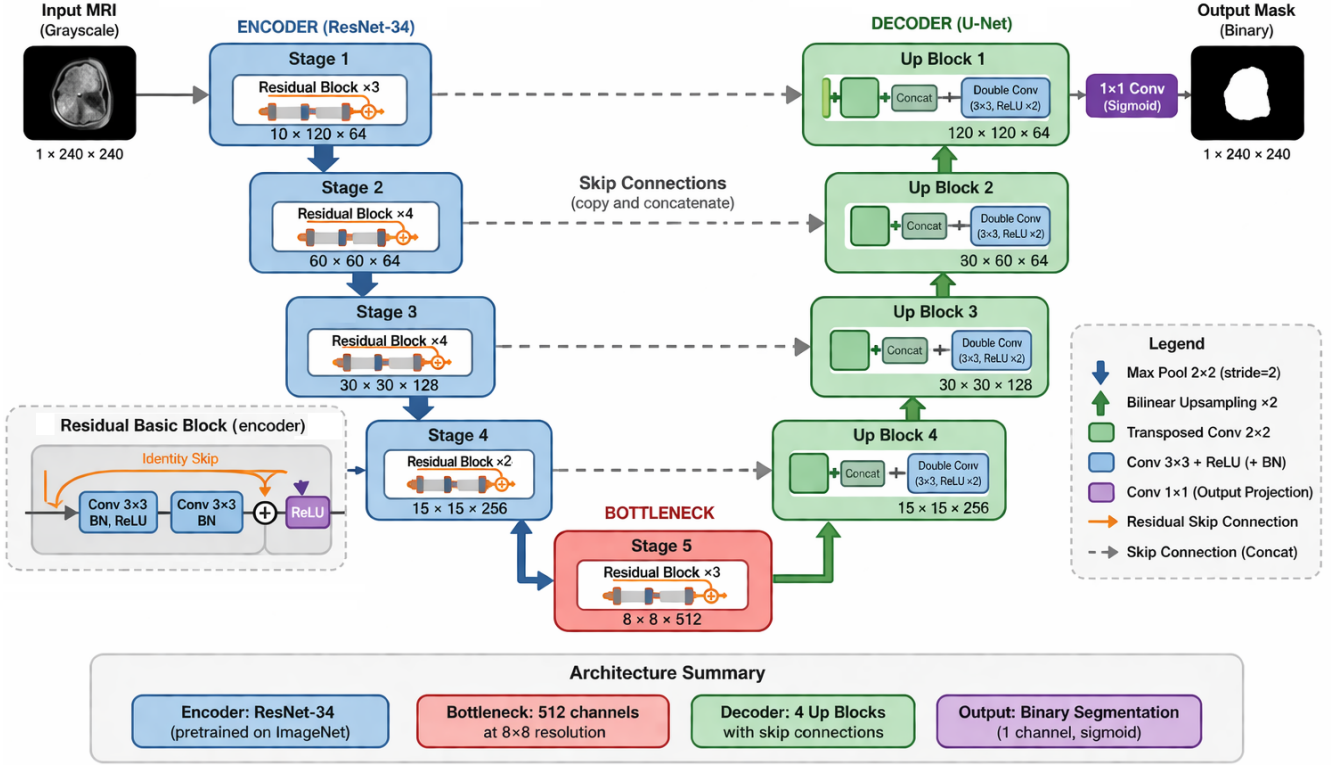


Fig. 3: Overview of the implemented 2D ResUNet architecture.

while IoU (Jaccard Index) is given by:

$$IoU = \frac{TP}{TP + FP + FN}. \quad (2)$$

Metrics were computed over all tumor-containing slices, and results are reported as the mean on the test sets, enabling consistent comparison across modalities and experimental scenarios.

A. Generalization Evaluation (RQ1/RQ2)

To rigorously assess cross-domain robustness, we design evaluation protocols that progressively increase the distributional discrepancy between training and testing data, enabling a controlled analysis of generalization behavior across MRI modalities.

Baseline. In the baseline setting, models are trained and evaluated within the same domain (BraTS 2020 or BraTS 2024 independently), providing an upper-bound reference under in-distribution conditions. Since training, validation, and test sets share the same underlying data distribution, this scenario isolates the intrinsic segmentation capability of each modality without domain shift effects.

Cross-Dataset. To explicitly evaluate generalization under domain shift, models trained on one dataset and directly tested on the other without adaptation. Two configurations are considered: BraTS 2024→2020 and BraTS 2020→2024. This protocol introduces discrepancies in acquisition protocols, intensity distributions, and population characteristics, providing a realistic assessment of performance degradation

across domains. It enables quantifying modality-specific robustness and identifying which MRI sequences are inherently more resilient to distributional shifts [6].

Multi-Domain Training. In this setting, training data from BraTS 2020 and BraTS 2024 are jointly used in balanced proportions (50/50), while preserving patient-level holdout splits to avoid data leakage. This protocol investigates whether exposure to heterogeneous data distributions improves generalization by promoting the learning of domain-invariant representations, and assesses the extent to which shared features across domains can mitigate performance degradation.

TABLE I: Dice/IoU (%) performance across evaluation protocols and MRI modalities.

Scenario	T1C	T1N	T2W	T2F
Baseline 2020	61.3 / 51.2	62.9 / 52.9	73.3 / 64.2	77.3 / 68.4
Baseline 2024	66.1 / 54.7	67.3 / 56.0	74.2 / 63.5	78.3 / 68.6
Cross-Dataset 20/24	49.7 / 38.8	49.7 / 39.1	62.3 / 51.4	58.2 / 46.9
Cross-Dataset 24/20	65.3 / 55.5	58.3 / 48.8	70.4 / 61.2	76.2 / 67.6
Multi-Domain 20/24	63.5 / 52.3	63.9 / 52.9	72.1 / 61.7	75.0 / 65.2

The results summarized in Table I provide key insights into cross-dataset generalization across MRI modalities. A consistent gap between in-domain and cross-domain performance is observed, confirming the sensitivity of segmentation models to domain shift (RQ1). While baseline results establish an upper bound, with FLAIR (T2F) and T2W achieving the highest Dice scores (e.g., 77.3–78.3

and 73.3–74.2), cross-dataset evaluation reveals substantial performance degradation, particularly for T2F (77.3→58.2) and T2W (73.3→62.3). Despite this, T2F maintains the highest cross-domain performance (up to 76.2), followed by T2W, whereas T1-based modalities show lower robustness (e.g., ≈ 49.7 Dice).

From a representation learning perspective, T2-based modalities enable more transferable features, while T1-based inputs induce more domain-dependent representations. The smaller drop (e.g., T2F: 78.3→76.2) suggests that training on BraTS 2024 yields more generalizable representations, driven by its larger scale and greater data diversity. Multi-domain training further mitigates degradation (e.g., T2F: 75.0 vs. 77.3/78.3), indicating that exposure to heterogeneous data promotes domain-invariant features. Overall, modality choice plays a central role in model generalization, with FLAIR providing the most robust representations under domain shift (RQ2).

B. Domain Adaptation (RQ3)

This scenario evaluates domain adaptation under limited labeled target-domain data by fine-tuning models, initially trained on a source dataset, using incremental fractions (10% steps) of annotated samples from the best-performing modality (T2F/FLAIR). Results show that fine-tuning with only 10–20% of target-domain data already approaches the upper-bound performance, after which a plateau is observed, indicating diminishing returns. This directly answers RQ3, demonstrating that a small amount of target data is sufficient to substantially improve generalization while reducing annotation and computational costs. Notably, the largest gains are observed when adapting models trained on BraTS 2020 to BraTS 2024 (Figure 4), whereas the reverse direction (Figure 5) yields more modest improvements ($\approx 5\%$ Dice), consistent with the generalization analysis in Section IV-A, which suggests that models trained on BraTS 2024 already learn more transferable representations.

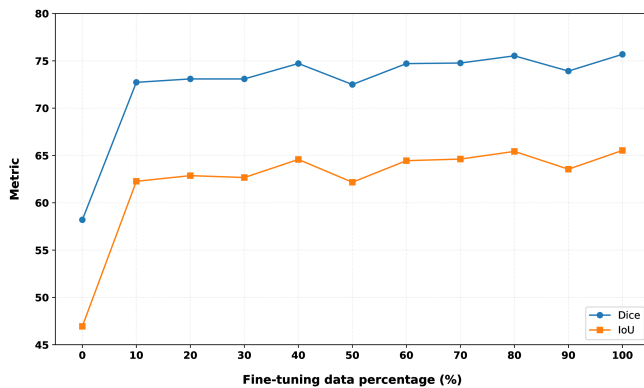


Fig. 4: Domain adaptation performance of a model trained on BraTS 2024 and progressively fine-tuned with increasing fractions of BraTS 2020 data.

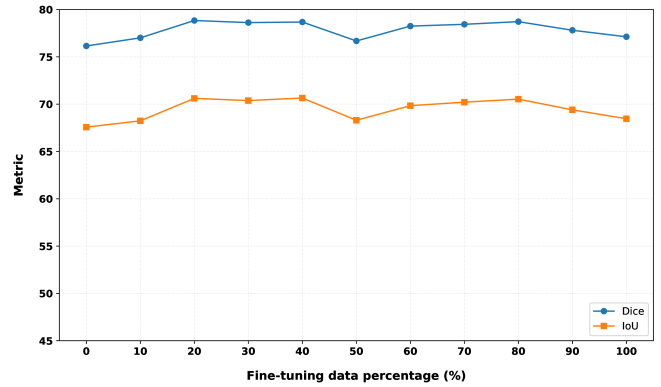


Fig. 5: Domain adaptation performance of a model trained on BraTS 2020 and progressively fine-tuned with increasing fractions of BraTS 2024 data.

C. Qualitative Analysis

Table II shows performance stratified by tumor size quartiles. Dice and IoU increase consistently from Q1 to Q4, reflecting the stronger penalization of small structures and the improved availability of morphological cues in larger lesions.

Modality performance also varies with size. For small tumors (Q1), T2f/FLAIR provides the best results, indicating higher sensitivity to subtle boundaries. For medium sizes (Q2–Q3), T2f/FLAIR and T2w achieve comparable performance, both outperforming T1-based modalities. For large tumors (Q4), T2w slightly surpasses T2f/FLAIR. Overall, T2f/FLAIR is preferable for small lesions, while T2f/FLAIR and T2w offer similar performance for larger volumes, with a slight advantage for T2w in the upper quartile.

TABLE II: Dice/IoU (%) results by size quartile.

Quartile / Size	T1C	T1N	T2W	T2F
Q1 (171 px)	26.9 / 19.6	26.0 / 18.8	39.5 / 30.3	49.0 / 39.2
Q2 (729 px)	63.3 / 51.1	64.0 / 51.4	75.3 / 63.8	77.6 / 66.8
Q3 (1475 px)	75.8 / 64.5	76.6 / 65.0	84.0 / 74.1	84.0 / 75.0
Q4 (2957 px)	79.7 / 68.6	81.3 / 70.1	85.7 / 76.4	84.8 / 76.0

To complement Table II and better illustrate performance across tumor sizes, Figure 6 presents segmentation masks for the median case of each quartile.

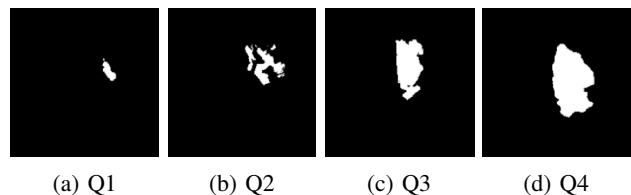


Fig. 6: Representative masks for each tumor size quartile.

To contextualize the quantitative results, we present qualitative examples on the T2f/FLAIR sequence, including correct, intermediate, and failure cases (Figures 7, 8, 9). Each panel shows the input image, ground-truth mask, input

overlaid with the ground truth (green), predicted mask, input overlaid with the prediction (red), and the final overlay, where correctly segmented regions (overlap between ground truth and prediction) are highlighted in yellow.

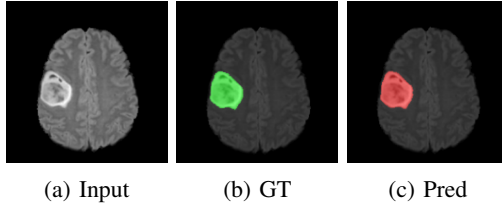


Fig. 7: Correct segmentation example (Dice: 99.1%).

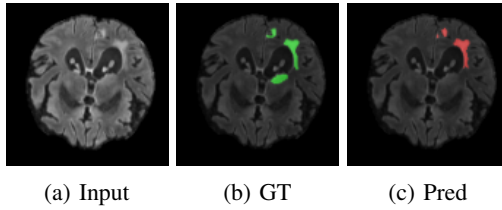


Fig. 8: Intermediate segmentation example (Dice: 70.1%).

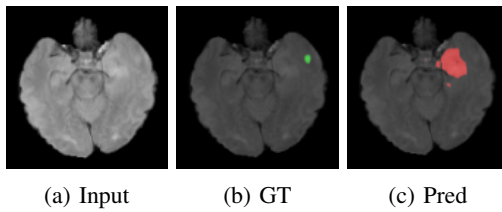


Fig. 9: Failure case (Dice: 0.0%).

V. CONCLUSION

This work investigated the role of MRI modalities in cross-dataset generalization through a standardized ResUNet-based pipeline and controlled evaluation protocols. The results support four main conclusions. First, T2f/FLAIR consistently provides the best performance and strongest generalization, followed closely by T2w, while T1-based modalities exhibit lower robustness under domain shift (RQ1/RQ2). Second, multi-domain training reduces the generalization gap, improving transfer performance and bringing results closer to in-domain conditions. Third, tumor size significantly affects performance: T2f/FLAIR is more effective for small lesions, whereas T2w becomes comparable or slightly superior for larger tumors. Fourth (RQ3), domain adaptation via fine-tuning is highly data-efficient, with 10–20% of target-domain data already approaching upper-bound performance and limited gains thereafter.

Overall, these findings demonstrate that model generalization is strongly influenced by modality choice, data diversity, and lesion characteristics, providing practical guidance for resource-constrained scenarios and modality selection in real-world applications.

Future work includes extending the analysis to multimodal configurations, exploring 3D architectures for improved volumetric consistency, developing strategies for small lesion detection, validating beyond the BraTS datasets, and assessing foundation models in terms of generalization, sensitivity, and computational cost.

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